

Sec 2 Math Cheat Sheet

Every formula, rule and method from the Sec 2 syllabus, grouped by the three MOE strands. Each topic has a quick worked example and the mistake that loses the most marks.

Numbers & Algebra 9 topics

Proportion

N&A

TYPE	EQUATION	GRAPH
Direct, $y \propto x$	$y = kx$	straight line through origin
Inverse, $y \propto \frac{1}{x}$	$y = \frac{k}{x}$	curve, never touches axes

Also $y = kx^2$, $y = k\sqrt{x}$, $y = \frac{k}{x^2}$. Direct: x doubles $\rightarrow y$ doubles.

Inverse: $xy = k$ is constant.

EXAMPLE

$y \propto x$, $y = 12$ when $x = 3$. So $k = 4$, and when $x = 5$, $y = 20$.

Watch out: always find k first, then substitute. Don't guess the ratio.

Linear graphs

N&A

$$y = mx + c$$

- m = gradient, c = y-intercept
- Gradient = $\frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$
- Positive m slopes up ↗, negative slopes down ↘
- Horizontal line $y = a$: gradient 0. Vertical line $x = a$: gradient undefined
- Parallel lines have the same gradient

EXAMPLE

Through (1, 3) and (4, 9): $m = 6 \div 3 = 2$. $3 = 2(1) + c \rightarrow c = 1$, so $y = 2x + 1$.

Watch out: rearrange into $y = \dots$ before reading m . In $2y = 6x - 4$, the gradient is 3, not 6.

Expansion

N&A

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(a + b)(a - b) = a^2 - b^2$$

General: $(a + b)(c + d) = ac + ad + bc + bd$ (multiply every term by every term).

EXAMPLE

$$(2x - 3)^2 = 4x^2 - 12x + 9$$

Watch out: $(a + b)^2 \neq a^2 + b^2$. Don't forget the middle term $2ab$. Also watch the sign: $-(x - 4) = -x + 4$.

Factorisation

N&A

Try these in order:

- Common factor:** $6x^2 + 9x = 3x(2x + 3)$
- Special products:** $x^2 - 25 = (x + 5)(x - 5)$
- Grouping:** $ax + ay + bx + by = (a + b)(x + y)$
- Quadratic** $x^2 + bx + c$: find two numbers that multiply to c and add to b (multiplication frame or cross method)

EXAMPLE

$x^2 - x - 12$: $-4 \times 3 = -12$ and $-4 + 3 = -1$, so $(x - 4)(x + 3)$.

$$2x^2 + 7x + 3 = (2x + 1)(x + 3)$$

Watch out: factorise completely. $2x^2 - 18 = 2(x^2 - 9) = 2(x + 3)(x - 3)$. Check by expanding back.

Algebraic fractions

N&A

- **Simplify:** factorise top and bottom, then cancel common factors
- **Add / subtract:** use the LCM of the denominators
- **Multiply:** factorise, cancel, then multiply across
- **Divide:** flip the second fraction, then multiply

EXAMPLE

$$\frac{x^2 - 9}{x^2 + 3x} = \frac{(x+3)(x-3)}{x(x+3)} = \frac{x-3}{x}$$

Watch out: you can only cancel factors, never terms. $\frac{x+3}{x} \neq 3$.

Changing the subject

N&A

Undo operations in reverse order, doing the same thing to both sides. If the new subject appears twice, collect those terms on one side and factorise it out.

EXAMPLE

Make r the subject of $V = \pi r^2 h$:

$$r^2 = \frac{V}{\pi h} \rightarrow r = \sqrt{\frac{V}{\pi h}}$$

EXAMPLE

$$ax + b = cx \rightarrow x(a - c) = -b \rightarrow x = \frac{b}{c - a}$$

Quadratic equations & graphs

N&A

Solve by factorisation: rearrange to $\dots = 0$, factorise, then set each factor to 0.

$$(x - p)(x - q) = 0 \Rightarrow x = p \text{ or } x = q$$

Graph of $y = ax^2 + bx + c$:

- $a > 0$: u shape with a minimum point. $a < 0$: n shape with a maximum point
- y-intercept = c . x-intercepts = the roots
- Line of symmetry sits halfway between the roots: $x = \frac{p + q}{2}$

EXAMPLE

$$x^2 - 5x = -6 \rightarrow x^2 - 5x + 6 = 0 \rightarrow (x - 2)(x - 3) = 0 \rightarrow x = 2 \text{ or } 3$$

Watch out: never divide both sides by x . In $x^2 = 4x$ that loses the answer $x = 0$. Write $x(x - 4) = 0$.

Simultaneous equations

N&A

- **Elimination:** make one variable's coefficients equal, then add or subtract the equations
- **Substitution:** make one variable the subject, then substitute into the other equation
- **Graphical:** the solution is the point where the two lines intersect

EXAMPLE

$$x + y = 10 \dots (1) \quad x - y = 4 \dots (2)$$

$$(1) + (2): 2x = 14 \rightarrow x = 7, \text{ then } y = 3.$$

Watch out: check both values in the other equation. Parallel lines give no solution.

Linear inequalities

N&A

Solve like an equation, with one extra rule:

$\times$ or $\div$ by a negative $\Rightarrow$ flip the sign



Filled dot ● for $\leq$ or $\geq$. Open dot ○ for $<$ or $>$.

EXAMPLE

$$5 - 2x > 11 \rightarrow -2x > 6 \rightarrow x < -3$$

Watch out: "largest integer x " for $x < -3$ is -4 , not -3 .

Geometry & Measurement 5 topics

Congruence & similarity

G&M

Congruent (same shape, same size). Tests for triangles:

SSS · SAS · AAS (ASA) · RHS (RHS: right angle, hypotenuse, side)

Similar (same shape, different size): corresponding angles equal and corresponding sides in the same ratio.

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = k$$

Two triangles with two pairs of equal angles are similar (AA).

Watch out: name vertices in matching order. $\triangle ABC \equiv \triangle PQR$ means $A \leftrightarrow P$, $B \leftrightarrow Q$, $C \leftrightarrow R$. SSA and AAA do not prove congruence.

Scale drawings & maps

G&M

Map scale 1 : n means 1 cm on the map is n cm on the ground.

$$\text{Area scale} = (\text{length scale})^2$$

EXAMPLE

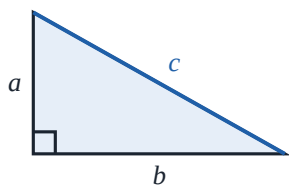
Scale 1 : 50 000 $\rightarrow$ 1 cm : 0.5 km, so 1 cm² : 0.25 km².
A 6 cm² park is $6 \times 0.25 = 1.5$ km².

UNIT CONVERSION

UNIT	CONVERSION
1 km	100 000 cm
1 m ²	10 000 cm ²
1 km ²	1 000 000 m ²
1 m ³	1 000 000 cm ³ = 1000 ℓ
1 cm ³	1 mℓ

Pythagoras' theorem

G&M



$$a^2 + b^2 = c^2$$

c is the hypotenuse, the longest side, opposite the right angle.

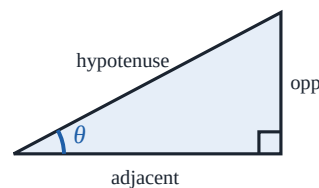
Converse: if $a^2 + b^2 = c^2$, the triangle is right-angled.

Common triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25 (and their multiples).

Watch out: finding a shorter side means subtracting: $a = \sqrt{c^2 - b^2}$.

Trigonometric ratios

G&M



$$\sin \theta = \frac{\text{opp}}{\text{hyp}} \quad \cos \theta = \frac{\text{adj}}{\text{hyp}} \quad \tan \theta = \frac{\text{opp}}{\text{adj}}$$

Remember **TOA CAH SOH**. Use $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ to find an angle.

EXAMPLE

hyp = 10, $\theta = 30^\circ$: opp = $10 \sin 30^\circ = 5$.

Watch out: calculator must be in **DEG** mode. Label opp/adj from the angle you are using. They swap for the other angle.

Volume & surface area

G&M

SOLID	VOLUME	SURFACE AREA
Prism	base area $\times h$	sum of all faces
Cylinder	$\pi r^2 h$	$2\pi r h + 2\pi r^2$
Pyramid	$\frac{1}{3} \times \text{base area} \times h$	base + triangular faces
Cone	$\frac{1}{3} \pi r^2 h$	$\pi r l + \pi r^2$
Sphere	$\frac{4}{3} \pi r^3$	$4\pi r^2$
Hemisphere	$\frac{2}{3} \pi r^3$	$2\pi r^2$ curved, $3\pi r^2$ total

Cone slant height: $l = \sqrt{r^2 + h^2}$

Watch out: h is the perpendicular height, not the slant height. Check if the question wants curved or total surface area.

Statistics & Probability 3 topics

Statistical diagrams

S&P

- **Dot diagram:** one dot per value on a number line. Good for small data sets
- **Stem-and-leaf:** keeps every value. Always include a key, e.g. 3 | 4 means 34. Leaves in ascending order
- **Histogram:** for grouped continuous data. Bars touch, and equal class widths are used in Sec 2

STEM-AND-LEAF

2 | 3 5 8

3 | 0 4 4 9

4 | 1 6 Key: 3 | 4 = 34

Watch out: spot misleading graphs: axes not starting at 0, uneven scales, or 3-D pictures that exaggerate size.

Mean, median, mode

S&P

$$\text{Mean} = \frac{\text{sum of values}}{\text{number of values}} = \frac{\sum fx}{\sum f}$$

- **Median:** middle value after sorting. Position $\frac{n+1}{2}$.
If n is even, average the two middle values
- **Mode:** most frequent value (can be none, or more than one)
- Grouped data: use class mid-values for x . The answer is an estimate

EXAMPLE

3, 7, 7, 9, 14: mean = $40 \div 5 = 8$, median = 7, mode = 7.

Watch out: an extreme value (outlier) pulls the mean. The median is better then. Sort the data before finding the median.

Probability

S&P

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

- $0 \leq P(E) \leq 1$. Impossible = 0, certain = 1
- $P(\text{not } E) = 1 - P(E)$
- List the sample space (all outcomes) when unsure.
For two dice there are $6 \times 6 = 36$ outcomes

EXAMPLE

Bag of 3 red and 5 blue: $P(\text{red}) = \frac{3}{8}$, $P(\text{not red}) = \frac{5}{8}$.

Watch out: give answers as fractions in simplest form (or decimals), never as ratios like 3 : 8.

Based on the MOE O-Level Mathematics syllabus (4052) content for Secondary 2, G3 / Express level. G2 students cover most of these topics, though some (such as trigonometry and algebraic fractions) may be taught at a different point. Check with your school's scheme of work.